

MODULES TRACK

Module 1: Python Libraries for Machine Learning

- NumPy
- Pandas
- Matplotlib
- Assignment

ASSIGNMENT INCLUDED

Module 2: Data Preprocessing

- Handling Missing Data
- Encoding Categorical Variables
- Feature Scaling (Normalization & Standardization)

ASSIGNMENT INCLUDED

Module 3: Supervised Machine Learning

A. Regression Algorithms

- 1. Linear Regression
- 2. Multiple Linear Regression
- 3. Polynomial Regression
- 4. Decision Tree Regression
- 5. Random Forest Regression

PROJECT INCLUDED

B. Classification Algorithms

- 1. Logistic Regression
- 2. Support Vector Machine (SVM)
- 3. K-Nearest Neighbors (KNN)
- 4. Naive Bayes
- 5. Decision Tree Classifier
- 6. Random Forest Classifier

PROJECT INCLUDED

Module 4: Unsupervised Machine Learning

- 1. K-Means Clustering
- 2. Hierarchical Clustering

PROJECT INCLUDED

Module 5: Deep Learning Fundamentals

- Neural Networks
- Activation Functions
- Optimizers
- Loss Functions & Backpropagation

PROJECT INCLUDED

Module 6: Convolutional Neural Networks (CNN)

- • Convolution, Pooling, Flattening
- • CNN Architecture in TensorFlow/Keras
- • Data Augmentation
- • Transfer Learning (VGG, ResNet overview)

PROJECT INCLUDED

Module 7: Recurrent Neural Networks (RNN)

- • Sequential Data and Time Series
- • RNN, LSTM, GRU Architectures
- • Text and Sentiment Analysis

PROJECT INCLUDED

Module 8: Large Language Models (LLMs)

- • Transformers Overview (Encoder-Decoder Architecture)
- • BERT, GPT, LLAMA, Gemini Overview
- • Prompt Engineering Basics
- • RAG (Retrieval-Augmented Generation)

PROJECT INCLUDED

Module 9: Model Deployment (Intro)

- • Introduction to Model Deployment
- • Streamlit Overview
- • AWS Overview